

Program : Diploma in Civil Engineering and Planning	
Course Code: 5399	Course Title: Computer Application Lab
Semester: 5	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To provide hands-on experience for the students with softwares in project management, structural analysis and design, concrete mix design, and estimating.
- To enable the students to do the practical problems by using the available application software packages.

Course Prerequisites:

Topic	Course code	Course name	Semester
Building Estimate		Estimating and costing	4
Structural analysis		Theory of structures	4
Mix design of concrete		Concrete technology	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Identify the task and duration and estimate the time required for completing a project using management software	11	Applying
CO2	Carry out the analysis and design of structures and prepare the detailing using a design software	12	Applying
CO3	Prepare the mix proportions for concrete with given strength and workability	9	Applying
CO4	Prepare the detailed estimate of a building using an estimation package	9	Applying
	Series test	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3		3	
CO2			3	3			
CO3		3		3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the task and duration and estimate the time required for completing a project using management software		
M1.01	Identify task & their duration	3	Applying
M1.02	Assign resources – Linking tasks	3	Understanding
M1.03	Estimate time required for completing the project	2	Applying
M1.04	Identify the critical path	3	Applying
CO2	Carry out the analysis and design of structures and prepare the detailing using design software		
M2.01	Perform analysis by using structural analysis and design application software (at least one package)	6	Applying
M2.02	Design of structures using the design package	3	Applying
M2.03	Preparation of ductile detailing by using any available package	3	Applying

	Series Test I	2	
CO3	Prepare the mix proportions for concrete with given strength and workability using any available Concrete Mix Design Package		
M3.01	Prepare the tables for material properties for mix design.	6	Applying
M3.02	Design the concrete mix for a given strength and workability using any available Concrete mix design package	3	Applying
CO4	Prepare a detailed estimate of a building using an estimation package		
M4.01	Perform detailed quantity calculations by using available software packages	5	Applying
M4.02	Prepare the conveyance statement with software packages	2	Applying
M4.03	Prepare detailed estimate and bill of quantities for a building by using available software packages	2	Applying
	Series Test II	2	

Text / Reference:

T/R	Book Title/Author
T1	Microsoft Project 2019 Step by Step Cindy Lewis, Carl Chatfield, Timothy Johnson
R1	Staad Pro V8i for Beginners: With Indian Examples Paperback by T S Sharma
R2	Practical Concrete Mix Design, Avijit Chaubey

Online Resources:

Sl.No	Website Link
1	https://resources.oreilly.com/examples/9780735626959-files/